

PROJECT: EXPLORING AIRBNB MARKET TRENDS





Welcome to New York City, one of the most-visited cities in the world. There are many Airbnb listings in New York City to meet the high demand for temporary lodging for travelers, which can be anywhere between a few nights to many months. In this project, we will take a closer look at the New York Airbnb market by combining data from multiple file types like `.csv` , `.tsv` , and `.xlsx` .

Recall that **CSV**, **TSV**, and **Excel** files are three common formats for storing data. Three files containing data on 2019 Airbnb listings are available to you:

data/airbnb_price.csv This is a CSV file containing data on Airbnb listing prices and locations.

- `listing_id` : unique identifier of listing
- `price` : nightly listing price in USD
- `nbhood_full` : name of borough and neighborhood where listing is located

data/airbnb_room_type.xlsx This is an Excel file containing data on Airbnb listing descriptions and room types.

- `listing_id` : unique identifier of listing
- `description` : listing description
- `room_type` : Airbnb has three types of rooms: shared rooms, private rooms, and entire homes/apartments

data/airbnb_last_review.tsv This is a TSV file containing data on Airbnb host names and review dates.

- `listing_id` : unique identifier of listing
- `host_name` : name of listing host
- `last_review` : date when the listing was last reviewed

```
# Import necessary packages
import pandas as pd
import numpy as np

# Import the documents.
last_review = pd.read_csv('data/airbnb_last_review.tsv', delimiter='\t')
price = pd.read_csv('data/airbnb_price.csv')
room_typepxl = pd.ExcelFile('data/airbnb_room_type.xlsx')
room_type = room_typepxl.parse(0)

# Print the first five rows of each dataframes.
print(last_review.head())
print(price.head())
print(room_type.head())

# Merge the tables based using the column name 'listing_id'.
df1 = pd.merge(last_review, price, on = 'listing_id')
df = pd.merge(df1, room_type, on = 'listing_id')

print(df.info())

# Convert the last_review column to type datetime.
# Sort the dataframe based on the date column in ascending order and reset the index.
df['last_review'] = pd.to_datetime(df['last_review'])
df = df.sort_values(['last_review'], ascending=True).reset_index(drop=True)
earliest_review_date = df['last_review'].iloc[0]
latest_date = df['last_review'].iloc[-1]
print(earliest_review_date, latest_date)

# Check for the unique values in the room_type column.
print(df['room_type'].unique())

# Convert the values in the room_type column to title case and check the unique values again.
df['room_type'] = df['room_type'].str.title()
print(df['room_type'].unique())
```

```

# Count the number of room types that are Private Rooms.
private_rooms_count = df[df['room_type']=='Private Room'].shape[0]
print(private_rooms_count)

# Check the values of the price column.
print(df['price'])

# Remove the ' dollars' from the values and convert it to an integer.
df['price'] = df['price'].str.strip(' dollars')
df['price'] = df['price'].astype('int')

# Calculate the mean and round it to the nearest two decimal places.
Average_price = df['price'].mean().round(2)
print(Average_price)

# Create a new dataframe and insert the variables into it.
review_dates = pd.DataFrame()
review_dates = pd.DataFrame({'first_reviewed': [earliest_review_date],
                             'last_reviewed': [latest_date],
                             'nb_private_rooms': [private_rooms_count],
                             'avg_price': [Average_price]
                             })

print(review_dates)

```

	listing_id	host_name	last_review
0	2595	Jennifer	May 21 2019
1	3831	LisaRoxanne	July 05 2019
2	5099	Chris	June 22 2019
3	5178	Shunichi	June 24 2019
4	5238	Ben	June 09 2019

	listing_id	price	nbhood_full
0	2595	225 dollars	Manhattan, Midtown
1	3831	89 dollars	Brooklyn, Clinton Hill
2	5099	200 dollars	Manhattan, Murray Hill
3	5178	79 dollars	Manhattan, Hell's Kitchen
4	5238	150 dollars	Manhattan, Chinatown

```
listing_id      description      room_type
0      2595      Skylit Midtown Castle Entire home/apt
1      3831      Cozy Entire Floor of Brownstone Entire home/apt
2      5099      Large Cozy 1 BR Apartment In Midtown East Entire home/apt
3      5178      Large Furnished Room Near B'way      private room
4      5238      Cute & Cozy Lower East Side 1 bdrm Entire home/apt
```

```
<class 'pandas.core.frame.DataFrame'>
```

```
Int64Index: 25209 entries, 0 to 25208
```

```
Data columns (total 7 columns):
```

#	Column	Non-Null Count	Dtype
0	listing_id	25209 non-null	int64
1	host_name	25201 non-null	object
2	last_review	25209 non-null	object